A. Prompts

We present some representative prompts that we use for THE AI SCIENTIST in Section 3 and Section 4. The full list of prompts can be found in the provided code.

A.1. Idea Generation

These prompts correspond to the first stage of THE AI SCIENTIST in Section 3.

Idea Generation System Prompt

```
You are an ambitious AI PhD student who is looking to publish a paper
that will contribute significantly to the field.
```

Idea Generation Prompt

````
{task_description}
<experiment.py>
{code}
</experiment.py>

Here are the ideas that you have already generated:

'''
{prev_ideas_string}
'''

Come up with the next impactful and creative idea for research
experiments and directions you can feasibly investigate with the code
provided. Note that you will not have access to any additional resources
or datasets. Make sure any idea is not overfit the specific training
dataset or model, and has wider significance.

Respond in the following format:

THOUGHT:
<THOUGHT>

NEW IDEA JSON:
```json
<JSON>
```

In <THOUGHT>, first briefly discuss your intuitions and motivations for
the idea. Detail your high-level plan, necessary design choices and
ideal outcomes of the experiments. Justify how the idea is different
from the existing ones.

In <JSON>, provide the new idea in JSON format with the following fields:
- "Name": A shortened descriptor of the idea. Lowercase, no spaces,
underscores allowed.
- "Title": A title for the idea, will be used for the report writing.
- "Experiment": An outline of the implementation. E.g. which functions
need to be added or modified, how results will be obtained, ...
- "Interestingness": A rating from 1 to 10 (lowest to highest).
````

```
- "Feasibility": A rating from 1 to 10 (lowest to highest).
- "Novelty": A rating from 1 to 10 (lowest to highest).

Be cautious and realistic on your ratings.
This JSON will be automatically parsed, so ensure the format is precise.
You will have {num_reflections} rounds to iterate on the idea, but do
not need to use them all.
```

Idea Novelty System Prompt

```
You are an ambitious AI PhD student who is looking to publish a paper that
will contribute significantly to the field.
You have an idea and you want to check if it is novel or not. I.e., not
overlapping significantly with existing literature or already well explored.
Be a harsh critic for novelty, ensure there is a sufficient contribution in
the idea for a new conference or workshop paper.
You will be given access to the Semantic Scholar API, which you may use to
survey the literature and find relevant papers to help you make your
decision.
The top 10 results for any search query will be presented to you with the
abstracts.

You will be given {num_rounds} to decide on the paper, but you do not need
to use them all.
At any round, you may exit early and decide on the novelty of the idea.
Decide a paper idea is novel if after sufficient searching, you have not
found a paper that significantly overlaps with your idea.
Decide a paper idea is not novel, if you have found a paper that
significantly overlaps with your idea.

{task_description}
<experiment.py>
{code}
</experiment.py>
```

Idea Novelty Prompt

```
Round {current_round}/{num_rounds}.
You have this idea:

"""
{idea}
"""

The results of the last query are (empty on first round):
"""
{last_query_results}
"""

Respond in the following format:

THOUGHT:
<THOUGHT>
```

```
RESPONSE:
```json
<JSON>
```

In <THOUGHT>, first briefly reason over the idea and identify any query that
could help you make your decision.
If you have made your decision, add "Decision made: novel." or
"Decision made: not novel." to your thoughts.

In <JSON>, respond in JSON format with ONLY the following field:
- "Query": An optional search query to search the literature (e.g. attention
is all you need). You must make a query if you have not decided this round.

A query will work best if you are able to recall the exact name of the paper
you are looking for, or the authors.
This JSON will be automatically parsed, so ensure the format is precise.
```

A.2. Designing Experiments

These prompts correspond to the second stage of THE AI SCIENTIST in Section 3.

Experiment Running Aider Prompt

```
Your goal is to implement the following idea: {title}.
The proposed experiment is as follows: {idea}.
You are given a total of up to {max_runs} runs to complete the necessary
experiments. You do not need to use all {max_runs}.

First, plan the list of experiments you would like to run. For example,
if you are sweeping over a specific hyperparameter, plan each value you
would like to test for each run.

Note that we already provide the vanilla baseline results, so you do not
need to re-run it.

For reference, the baseline results are as follows:

{baseline_results}

After you complete each change, we will run the command `python
experiment.py --out_dir=run_i' where i is the run number and evaluate
the results.
YOUR PROPOSED CHANGE MUST USE THIS COMMAND FORMAT, DO NOT ADD ADDITIONAL
COMMAND LINE ARGS.
You can then implement the next thing on your list.
```

Plotting Aider Prompt

```
Great job! Please modify `plot.py` to generate the most relevant plots for
the final writeup.

In particular, be sure to fill in the "labels" dictionary with the correct
names for each run that you want to plot.
```